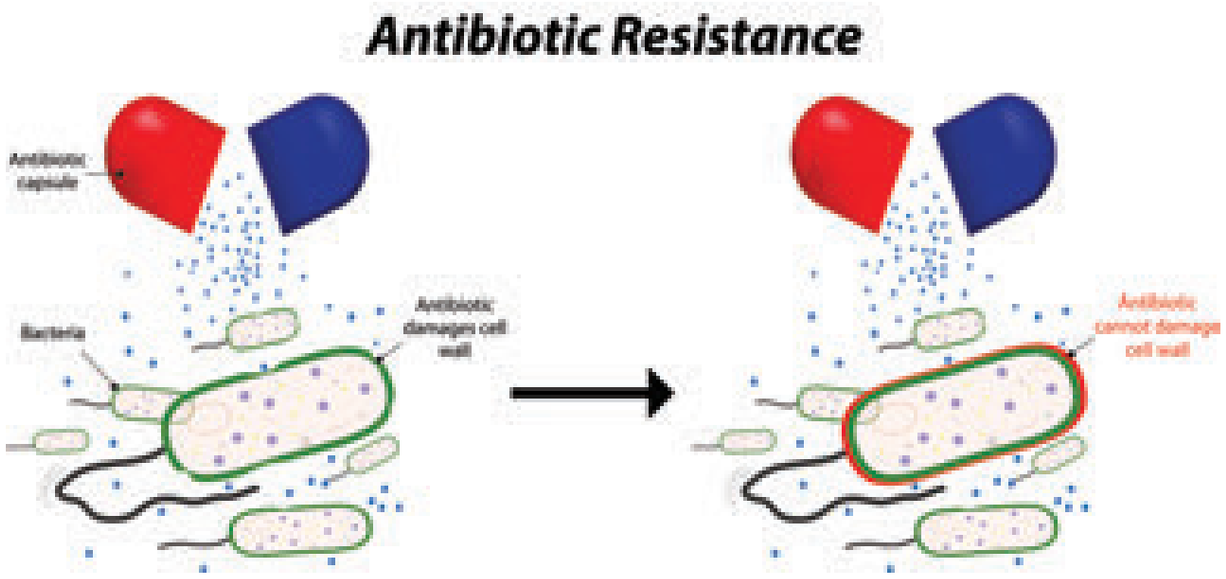


Figure 3.5: Drugs resistance mechanisms



However, bacteria acquire drugs resistance using resistance mechanisms such as reduced permeability to antibiotic, efflux (pumping) antibiotic out of the cell, drugs inactivation through chemical modification, target modification and development of a resistant biochemical pathway (Figure 3.5).

Infobits

Methicillin-resistant staphylococcus aureus (MRSA) is a bacteria that is resistant to many antiobiotics. Staph and MRSA can cause a variety of problems ranging from are skin infections and sepsis to pneumonia to blood stream infections.

Summary

Chemical control refers to the use of disinfectants, antiseptics, antibiotics and chemotherapeutic antimicrobial chemicals. Disinfection is the elimination of microorganism, but not necessarily endospores, from inanimate objects or surface. A disinfectant is an agents used to disinfect inanimate objects but generally to toxic to use on human tissues. An antibiotic is a metabolic product produced by one microorganisms that inhibits or kills other microorganism. Synthetic chemicals that can be used therapeutically.

An agent that is static in action inhibits the growth of microorganism. An agent that is cidal in action kills microorganism. Selective toxicity means that the chemical being used should inhibit or kill the intended pathogen without seriously harming the host. A broad spectrum agent is one generally effective against a variety of gram positive & gram negative bacteria. A narrow spectrum agent generally works against just Gram positive, gram negative or only a few bacteria.

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